TP1-S1 : Problem → Algorithm → Program

TP Série 1 • Session 1

Module: Informatique 1 | Duration: 1h30 | L1 – Socle Commun Sciences de la Matière

Learning objectives

→ Distinguish a problem, an algorithm and a program.
→ Identify input, processing and output.
→ Write clear step-by-step algorithms.

Exercises

Exercise 1

Classify: Algorithm / Program /Problem:
(a) Calculate the average of three temperatures.
(b) Read T1, T2, T3; calculate S; calculate A=S/3; display A.
(c) Average=(T1+T2+T3)/3.
(d) Determine density from mass and volume.
(e) A sequence of precise steps used to solve a problem.

Exercise 2

A technician measures mass m (g) and volume V (mL). Density is ρ=m/V.
- State the problem, inputs, output, and write a step-by-step algorithm.

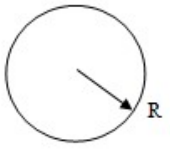
Exercise 3

Write an algorithm that reads Celsius C and calculates Fahrenheit F = (9/5) C + 32. Identify input, processing and output.

Exercise 4

A solution contains mass m (g) in volume V (L). C = m/V. Write the algorithm and identify input, processing and output.

Exercise 5



Write an algorithm that prompts the user to enter the radius R of a circle, then calculates and displays its perimeter P and area S.

- Formulas: $P = 2\pi R$ and $S = \pi R^2$

Exercise 6

Write an algorithm that reads the lengths of the three sides of a triangle *a, b, c*, and then calculates its area *S* using Heron's formula.

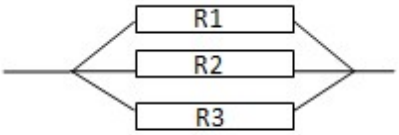
$$area = \sqrt{p(p-a)(p-b)(p-c)} \qquad and \qquad p = (a+b+c)/2$$

Exercise 7

Write an algorithm that calculates the kinetic energy E_c (in Joules) of an object of mass m (in kg) moving at a velocity v (in m/s).

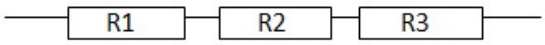
Exercise 8

Write an algorithm that prompts for the values of three resistors in Ohms and calculates the equivalent resistance:



When connected in **parallel**: : 1/Req = 1/R1 + 1/R2 + 1/R3

When connected in series: Req = R1 + R2 + R3



Exercise 9

Write an algorithm that calculates the kinetic energy E_C (in Joules) of an object of mass m (in kg) moving at a velocity v (in m/s).

Formula: $E_c = \frac{1}{2}mv^2$

Exercise 10

Write an algorithm and a Python program that reads the plate area A (in m^2), separation distance d (in m), relative permittivity ε_r, and applied potential difference V (in Volts). Calculate and display:

1. Capacitance (C in Farads, F): $C = \frac{\varepsilon_0 \varepsilon_r A}{d}$
2. Stored Electric Charge (Q in Coulombs, C): $Q = C \cdot V$
3. Electric Field Strength (E in V/m): $E = \frac{V}{d}$

(Constant: Permittivity of vacuum $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m)

Exercise 11

Complete: REAL-WORLD PROBLEM → _____ → _____. Explain what changes when an algorithm becomes a program.

Mini-challenge

Write an algorithm for the area of a rectangle from L and W.